REIMAGINING & REFORMING HEALTHCARE IN INDIA

VrikkaTwin

A kidney digital twin for heatwaves and sick days

Team Monarch
Harshita Kumar & Eeshan Gupta
Vellore Institute of Technology (VIT), Vellore

Happiest Health Digital Twin Challenge 2026 · "Vrikka" is Sanskrit for kidney



Wearables measure the heat.
Labs measure the kidney.
VrikkaTwin decides when to look.

VIRTUAL PATIENT DATA

AI INSIGHTS. OUTCOMES.

PREDICTIVE CARE

Meet Riya: one hot week, three ordinary choices

Riya (fictional), 68, Nagpur

Lives at home · high blood pressure · early kidney decline

- **Takes a BP pill** (ARB) and a water pill (diuretic) every day
- **Knee pain:** buys an ibuprofen painkiller (e.g. Combiflam) over the counter
- **May 2024:** Nagpur touches 45 °C, then she catches a fever

Every choice is normal. Together, they can injure her kidneys within days, and nobody is watching.

Day 1-6

Heat: she sweats, drinks too little

Day 7

Fever starts; painkiller 3× a day

Day 8

Kidney filtration quietly falls

Day 9

Acute kidney injury (creatinine up)

The problem

Why heat, sickness and common pills hurt Indian kidneys, and why nobody sees it coming

This is not rare: the people at risk number in crores

315 M

Indians with high blood pressure

ICMR-INDIAB, Lancet Diabetes Endocrinol 2023

138 M

Indians with chronic kidney disease, 2nd highest in the world

GBD 2023 (Lancet, 2025), via PTI report

67,637

suspected heatstroke cases and 374 deaths, 1 Mar - 27 Jul 2024

NCDC surveillance, Lok Sabha reply, Aug 2024

+82%

kidney-injury rate in the first 30 days of BP pill + water pill + painkiller

Lapi et al., BMJ 2013, 487,372 patients

And it is getting hotter: the ILO projects India will lose 5.8% of working hours to heat stress by 2030.

Why it happens: heat drains water, pills switch off the valves

1 · Water loss

Sweat in heat, fever, vomiting or loose motions drain body water. Older people feel less thirst.

2 · Safety valves switched off

Kidneys protect filtration with two 'valves'. Painkillers (NSAIDs) block the inlet valve; BP pills (ACE-inhibitors/ARBs) block the outlet; water pills remove more water.

3 · Low kidney reserve

Age 60+, chronic kidney disease, diabetes or heart failure leave less room for error.

Result: acute kidney injury (AKI) in 1-3 days, defined by KDIGO as a creatinine rise of ≥0.3 mg/dL in 48 h. Usually mild and reversible, but repeated injuries are linked to faster long-term decline.

Today: the danger lives in a window nobody watches

Minutes to hours

Heat alarms and heat-stress wearables

Warn about overheating, not about kidneys or medicines

1 to 3 days

Kidney injury from heat + sickness + pills

No routine tool watches this window

Months to years

Chronic kidney risk tests

Need a blood draw; ignore today's heat and illness

- **Kidney blood tests happen every 3-12 months**, so a 3-day injury is usually missed.
- **Heat alerts are the same for everyone**, not tied to a person's medicines or kidneys.
- **Painkillers are sold over the counter**, and Combiflam-type pills are a household habit.
- **Blanket 'sick-day rules' have weak evidence**; UK guidance supports individual assessment for high-risk patients (Think Kidneys, 2018). That individual assessment is exactly what a twin can do.

If nothing changes: one bad week can echo for years

What one missed injury can mean

- A hospital visit that a cheap action could have prevented: one pill swapped, ORS, a ₹189 creatinine test
- Any intercurrent AKI was associated with 1.83 mL/min/yr extra eGFR loss in an Indian diabetic kidney cohort (Trichy, n = 274)
- Faster progress toward dialysis, which many families cannot afford
- Doctors only find out at the next routine test, after the damage

Riya's slow clock (projection, 10-90% range)

Scenario	Median eGFR in 10 years
Continue as is	43
One stage-1 kidney injury this season	38
Start an SGLT2 inhibitor now	54

eGFR in mL/min/1.73 m². Ranges are wide (e.g. 9.6-75.5 for 'continue'); these are illustrative projections from published effect sizes, not prognosis. Source: reports/slowclock_demo.json.

Prevention is cheap. Knowing who to act on, today, is the missing piece.

Our answer: VrikkaTwin

A living virtual copy of each at-risk patient that decides when to look and what to change



The idea: exposure → organ → decision



Exposure

Wearables, home scale and weather measure what hits the kidney: heat, weight loss, heart rate, fever



Organ

Labs measure the kidney itself: creatinine and eGFR, but only every few months



Decision

The twin bridges the gap: who needs a test today, which pill to review, what to change

Fast clock · next 72 hours
Probability of acute kidney injury, with plain-language reasons and safer what-if choices

Slow clock · next 10 years
Projected kidney function as honest ranges, and what each decision does to it

Why this is a true digital twin, not just a risk score

Personalised

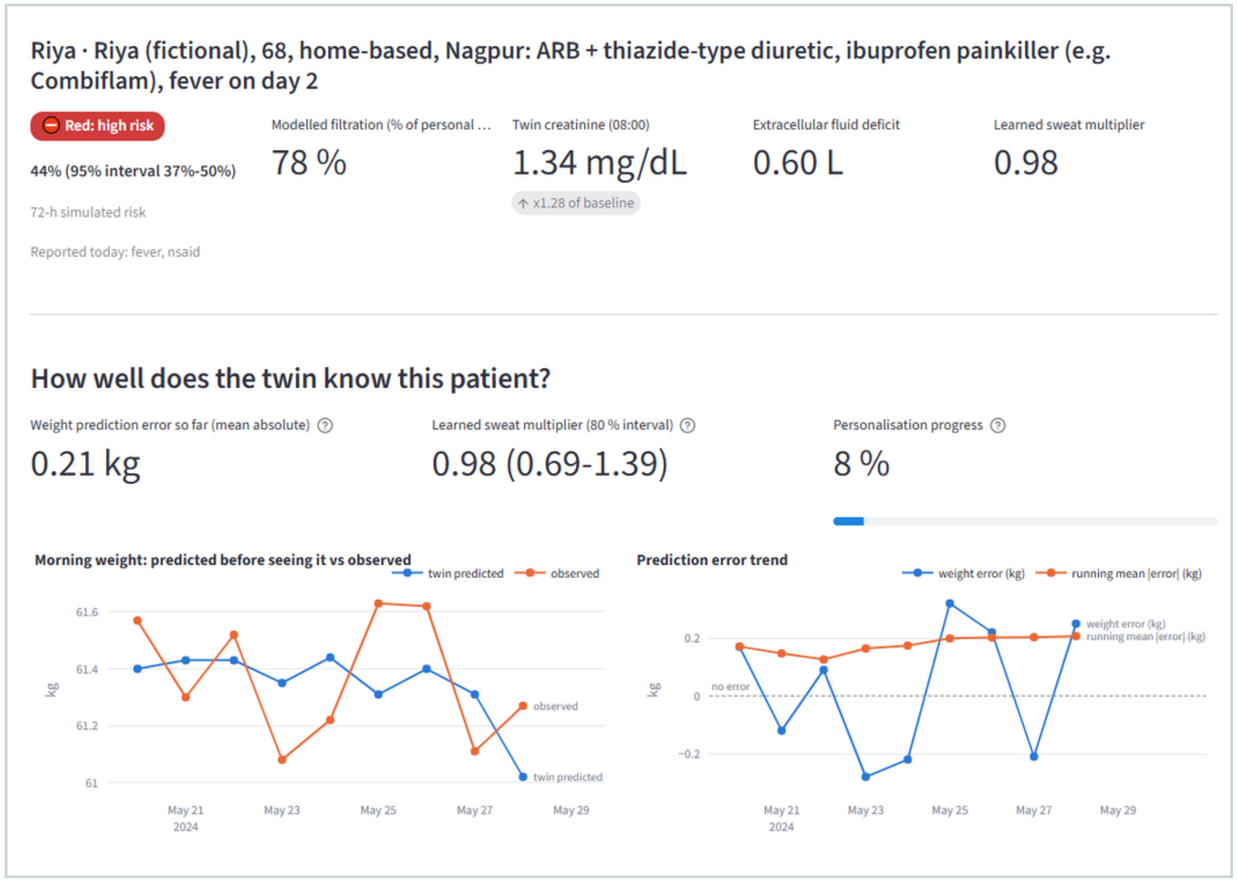
Each patient has their own hidden state: body water, creatinine, kidney reserve, learned sweat rate, CYP2C9 gene

Continuously synchronised

A 5-state Unscented Kalman Filter updates every morning from weight and heart rate, and resets when a lab result arrives

Simulatable

The doctor changes pills, fluids or work hours on the virtual patient and sees the 72-hour risk and 10-year range change



Live proof: the twin predicts Riya's morning weight before seeing it (mean error 0.21 kg, about the scale's noise) and is still learning her sweat rate (0.98, 80% interval 0.69-1.39).

Who it serves, even without a smartwatch

Doctor

Sorted risk list, reasons, what-if, one-click home test

Patient & family

Traffic light in their language with audio, and safe actions

ASHA worker

Tier-0 visit form, red-flag check, Telugu by default

Lab

Home-collection test orders; a result resets the twin

Tier 0 · No wearable

ASHA visit or phone check-in + free weather + clinic labs

≈ ₹0 extra

Tier 1 · Basic band + scale

Daily weight, resting heart rate, steps, sleep

band ≈ ₹1,500

Tier 2 · Richer devices

Home BP cuff, HRV, skin temperature

optional

A creatinine blood test costs about ₹189 (Apollo 24|7, Delhi listing). The twin rations tests: it decides who needs one today, instead of adding tests for everyone.

How it works

Data fusion, architecture, the physiology engine and a day in the life of the twin

Data fusion: static health record + live wearable stream

Static / historical (simulated EHR)

- Demographics: age, sex, weight, city, occupation, housing
- Past diagnoses: hypertension, diabetes, CKD stage, heart failure, prior AKI
- Lab results: creatinine, eGFR (CKD-EPI 2021), potassium, UACR
- Medicines: Indian brand names mapped to drug class (e.g. Telma-H, Combiflam)
- Genetic marker: CYP2C9 (*3 allele ≈ 0.11 in IndiGenomes) changes ibuprofen exposure

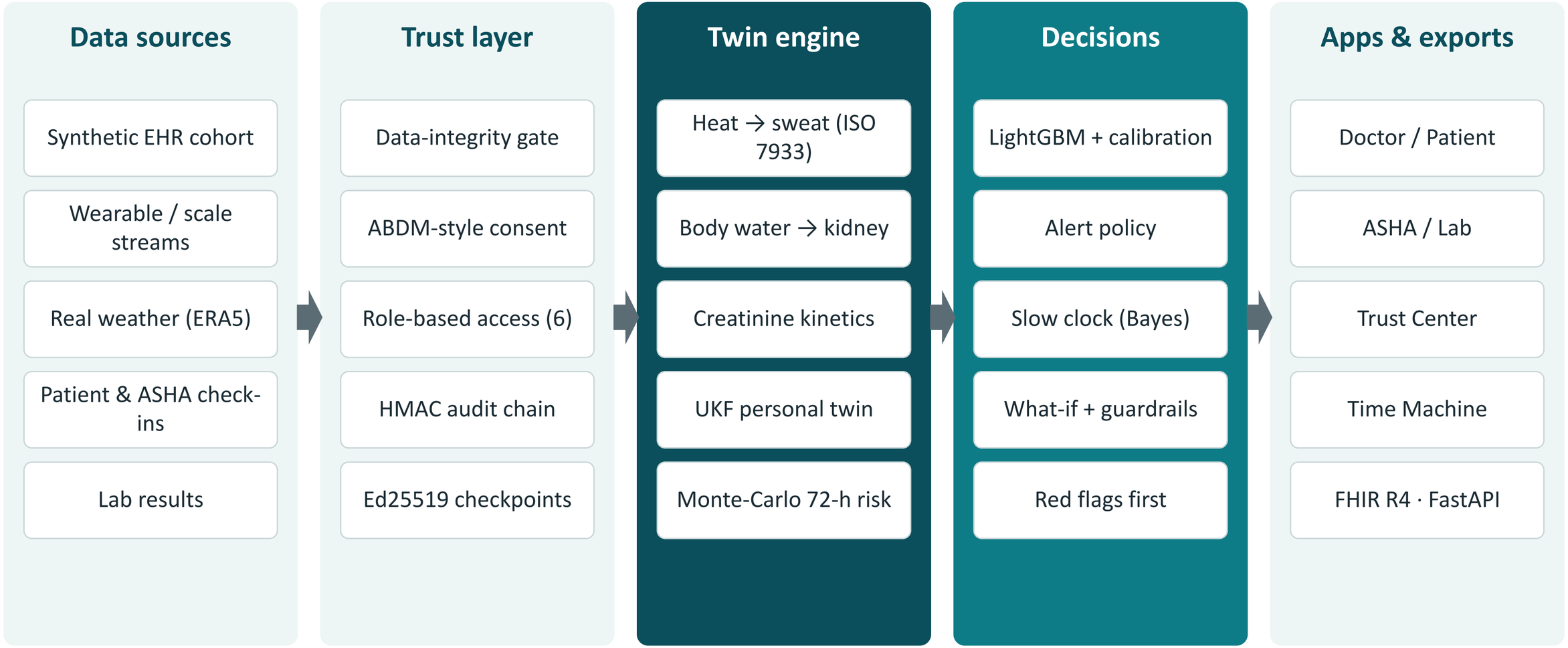
Twin

Dynamic / real-time (simulated wearables)

- Resting heart rate, HRV, steps, sleep (Fitbit / Apple Health-style)
- Morning weight from a home scale (water lost)
- Home BP (optional)
- Hourly weather: temperature, humidity, sun (real 2024-26 Open-Meteo / ERA5)
- Check-ins: fever, vomiting, tablets taken, ORS, work hours

Exactly the two streams the challenge asks for. Genotypes are simulated from published Indian allele frequencies; CPIC 2020 gives recommendations for ibuprofen, none for diclofenac or aceclofenac, so we apply no gene effect there.

Architecture



Python 3.12 · Streamlit UI calls one services layer that enforces role + consent + audit on every read; the same services sit behind FastAPI. Full diagram: deliverables/architecture.pdf.

Mechanism: from today's weather to tomorrow's kidney

Heat → sweat

ISO 7933 predicted-heat-strain model × the person's learned sweat rate

Two water deficits

Total water (seen by the scale) and extracellular water (felt by the kidney)

Kidney filtration

Autoregulation curve; painkiller, BP pill and water pill effects with real wear-off times

Creatinine

Mass-conserving kinetics, incl. haemoconcentration; read at 08:00 like a lab

72-h risk

200 Monte-Carlo futures with a noisy weather forecast; risk = share that meets KDIGO

Every morning the loop closes: the twin predicts today's weight and heart rate, compares them with the real readings, corrects its hidden state and learns the person's sweat rate. A lab result resets creatinine. Then a calibrated LightGBM model turns the twin's state into the alert.

A day with VrikkaTwin

1 · 07:00

Morning weight, resting HR and check-in arrive (or the ASHA visits)

2 · 08:00

The twin updates Riya's hidden state and creatinine estimate

3 · 09:00

Risk for the next 72 h; red flags are checked first

4 · Morning

Doctor sees the sorted list, the reasons and tomorrow's Kidney Weather

5 · Review

Doctor tries changes on the twin: painkiller swap, ORS, pauses

6 · Same day

One click orders a home kidney test; the lab result resets the twin

The product, screen by screen

Real screenshots from our working prototype (synthetic data, fictional personas)

Doctor Triage: who needs attention today, and why

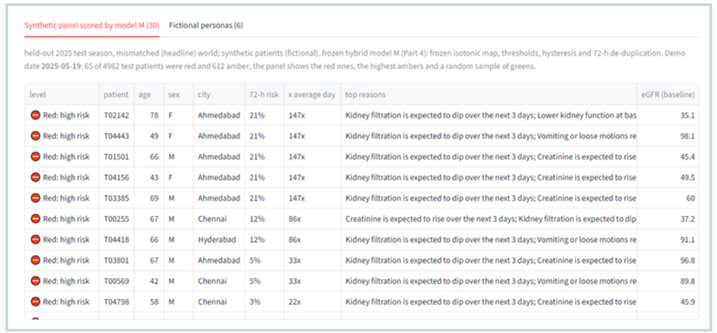
Doctor Triage

Tomorrow's Kidney Weather (2025-05-20)

Forecast: noisy 72-h forecast issued 09:00 on the demo day (the fast clock's input). Levels: today's level (its 72-hour window covers tomorrow), frozen model M, all held-out test patients (synthetic).

city	forecast max tomorrow (°C)	what ERA5 later recorded (°C)	patients	✅ Green: low risk	⚠️ Amber: risk is building	⛔ Red: high risk
Delhi	45.6	42.6	867	685	158	24
Ahmedabad	39.8	41.1	727	548	155	24
Nagpur	41.3	40.6	777	638	132	7
Hyderabad	36.7	34.5	682	613	63	6
Chennai	29.2	31.6	574	536	35	3
Palasa	32	31.5	605	550	54	1
Bengaluru	28.1	27.4	730	715	15	0

Suggested home kidney tests today (every patient in the red state today (red = doctor review + home kidney test, config/alert_policy.yaml), highest risk first; showing 20 of 65)



Synthetic panel scored by model M (30) | Fictional personas (6)

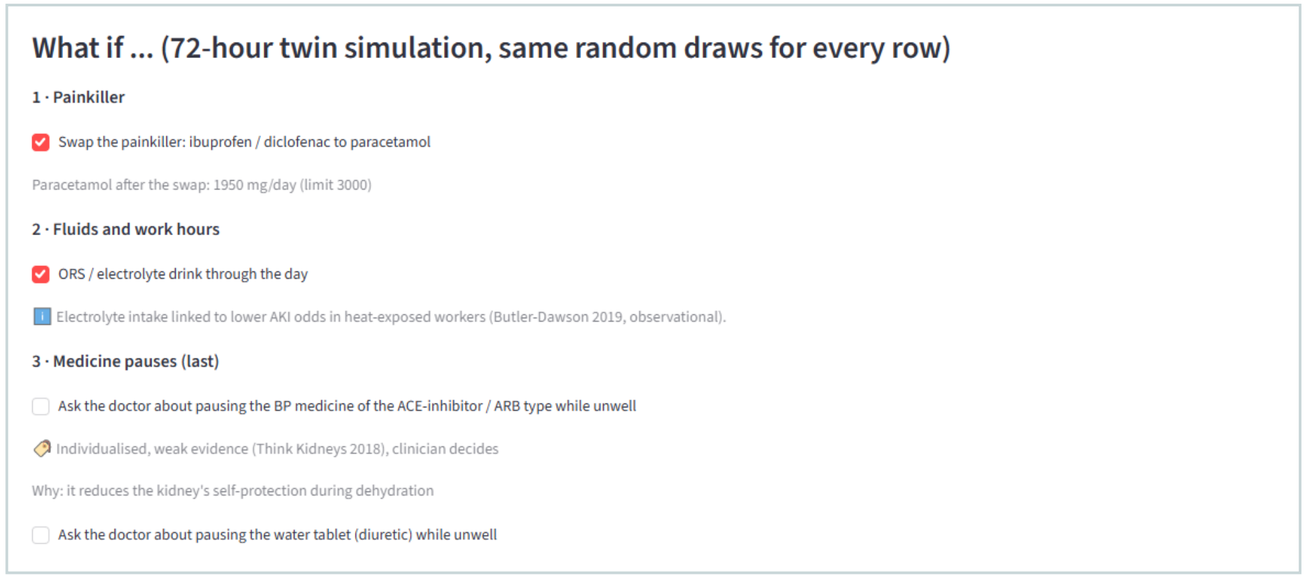
held-out 2025 test season, mismatched (headline) world; synthetic patients (fictional). frozen hybrid model M (Part 4): frozen isotonic map, thresholds, hysteresis and 72-h de-duplication. Demo date **2025-05-19**: 65 of 4962 test patients were red and 612 amber; the panel shows the red ones, the highest ambers and a random sample of greens.

level	patient	age	sex	city	72-h risk	x average day	top reasons	eGFR (baseline)
⛔ Red: high risk	T02142	78	F	Ahmedabad	21%	147x	Kidney filtration is expected to dip over the next 3 days; Lower kidney function at bas	35.1
⛔ Red: high risk	T04443	49	F	Ahmedabad	21%	147x	Kidney filtration is expected to dip over the next 3 days; Vomiting or loose motions re	98.1
⛔ Red: high risk	T01501	66	M	Ahmedabad	21%	147x	Kidney filtration is expected to dip over the next 3 days; Creatinine is expected to rise	45.4
⛔ Red: high risk	T04156	43	F	Ahmedabad	21%	147x	Kidney filtration is expected to dip over the next 3 days; Creatinine is expected to rise	49.5
⛔ Red: high risk	T03385	69	M	Ahmedabad	21%	147x	Kidney filtration is expected to dip over the next 3 days; Creatinine is expected to rise	60
⛔ Red: high risk	T00255	67	M	Chennai	12%	86x	Creatinine is expected to rise over the next 3 days; Kidney filtration is expected to dip	37.2
⛔ Red: high risk	T04418	66	M	Hyderabad	12%	86x	Kidney filtration is expected to dip over the next 3 days; Vomiting or loose motions re	91.1
⛔ Red: high risk	T03801	67	M	Ahmedabad	5%	33x	Kidney filtration is expected to dip over the next 3 days; Creatinine is expected to rise	96.8
⛔ Red: high risk	T00569	42	M	Chennai	5%	33x	Kidney filtration is expected to dip over the next 3 days; Vomiting or loose motions re	89.8
⛔ Red: high risk	T04798	58	M	Chennai	3%	22x	Kidney filtration is expected to dip over the next 3 days; Creatinine is expected to rise	45.9

- **Tomorrow's Kidney Weather:** forecast per city with green / amber / red counts, e.g. Delhi 45.6 °C: 685 / 158 / 24 of 867 patients
- **Suggested home kidney tests today:** every red patient, highest risk first, with the main reason
- **Never colour alone:** every level has an icon and a word
- **One click:** order a home test or download a FHIR R4 record

Held-out 2025 test season, mismatched simulator, frozen model M. Demo date 19 May 2025: 65 of 4,962 synthetic patients red, 612 amber.

Patient Twin: try a decision on the virtual patient first





One-click printable visit summary in the patient's language (Hindi shown), with the disclaimer.

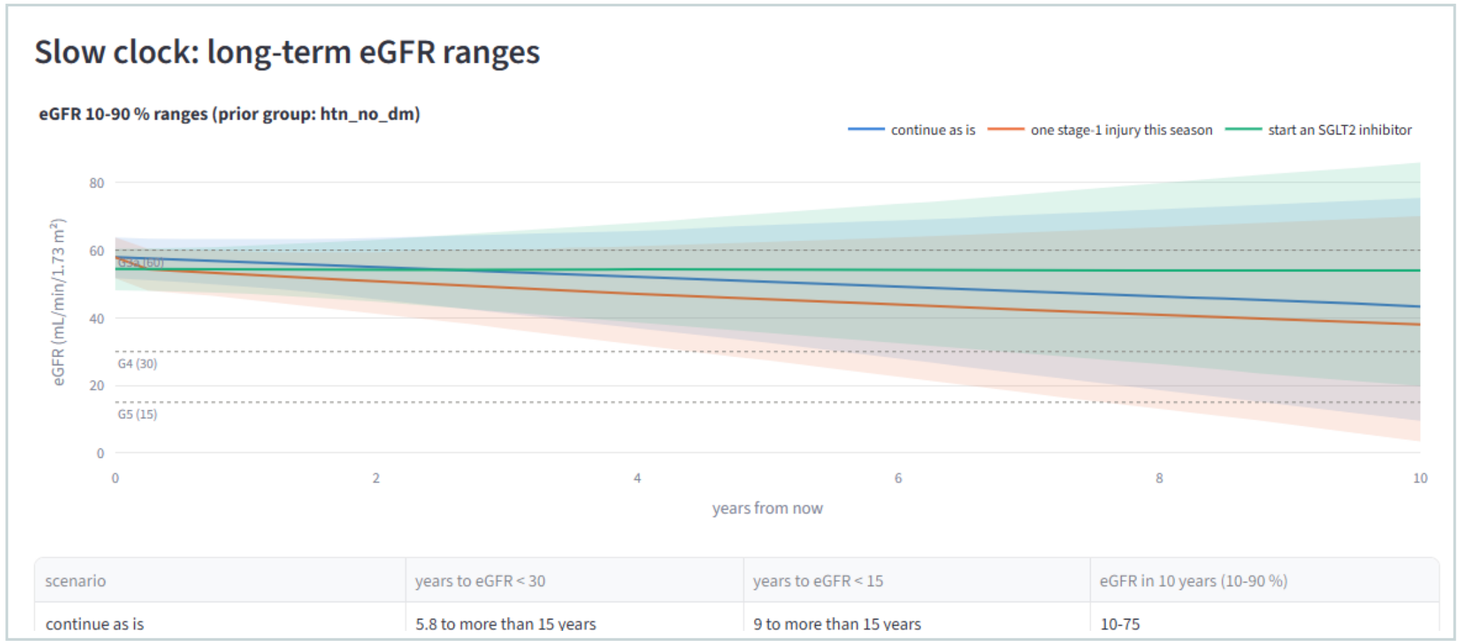
Riya, day 8 (72-hour risk, synthetic)

Continue as is	48%
1 · Swap ibuprofen → paracetamol	25.5%
2 · + ORS 1 L/day	1.5%
3 · + pause ARB & diuretic*	<1%
Same tablets, CYP2C9 poor metaboliser	55.5%

**Medicine pauses come last: individualised, weak evidence (Think Kidneys 2018), clinician decides. Guardrails block unsafe levers (heart failure, potassium ≥ 5.5, paracetamol ceiling 3 g/day if 65+).*

Source: reports/demo_numbers.json. The patient view never says 'stop your medicine'; it says 'talk to your doctor first'.

Slow clock: what this season means for the next 10 years



scenario	years to eGFR < 30	years to eGFR < 15	eGFR in 10 years (10-90 %)
continue as is	5.8 to more than 15 years	9 to more than 15 years	10-75

- **Bayesian eGFR slope** from the patient's own past labs and a group prior
- **Three scenarios:** continue, one stage-1 injury this season, start an SGLT2 inhibitor
- **SGLT2 effect from EMPA-KIDNEY:** 1.37 mL/min/yr slower decline (95% CI 1.16-1.59)
- **Always a 10-90% range,** labelled 'projection'; KFRE shown only where valid (eGFR 10-59 + UACR)

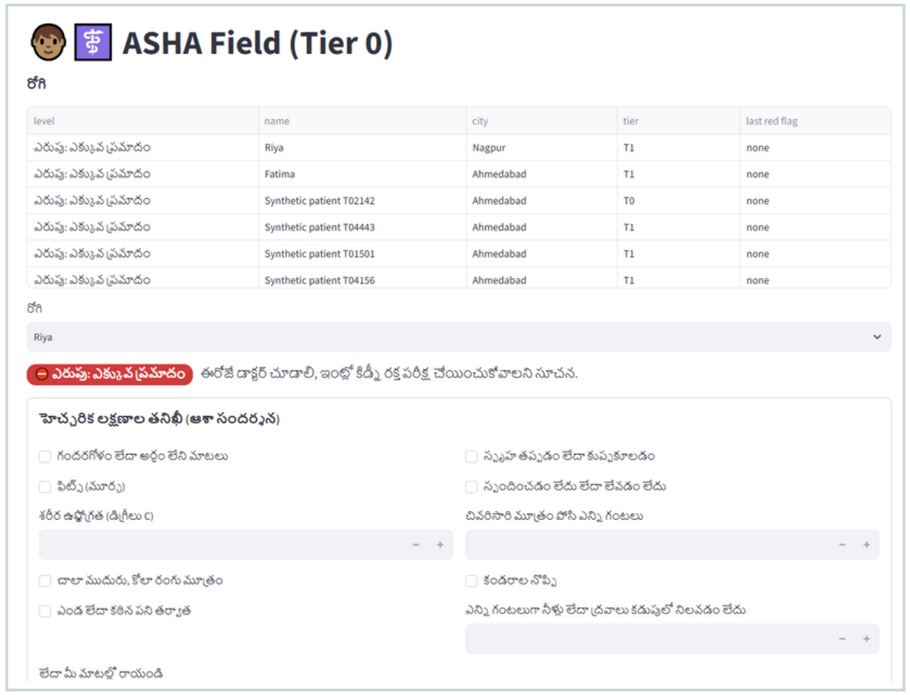
The AKI penalty fades over time and is 'associated with', from one Indian diabetic kidney disease cohort (n = 274); extrapolated for non-diabetics.

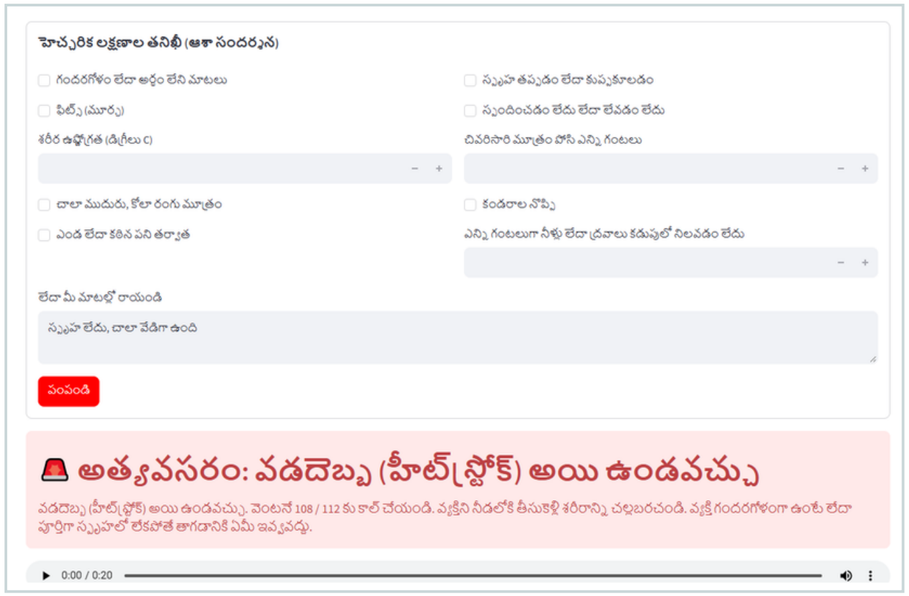
Patient & family: in their language, with their voice



- **4 languages:** English, हिन्दी, ಕನ್ನಡ, తెలుగు (Indian translations marked 'draft' until native-speaker review)
- **Listen button:** 28 pre-recorded spoken alerts work offline
- **Plain words,** never 'AKI'; red / amber / green with an icon and a word
- **Check-in with red flags first;** types Indian brand names, e.g. 'Combiflam' → ibuprofen + paracetamol
- **Patient controls consent:** stop sharing with the doctor, ASHA, lab or family in one tap

ASHA Tier 0 and red flags: safety before any score



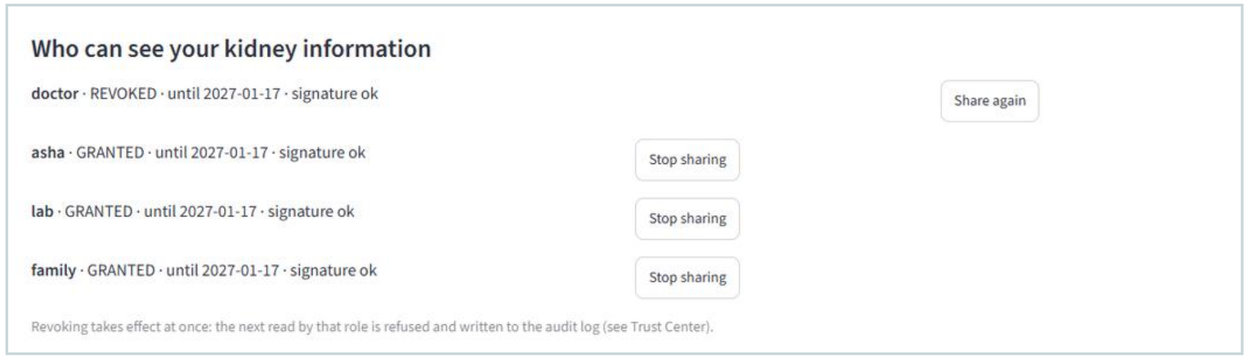


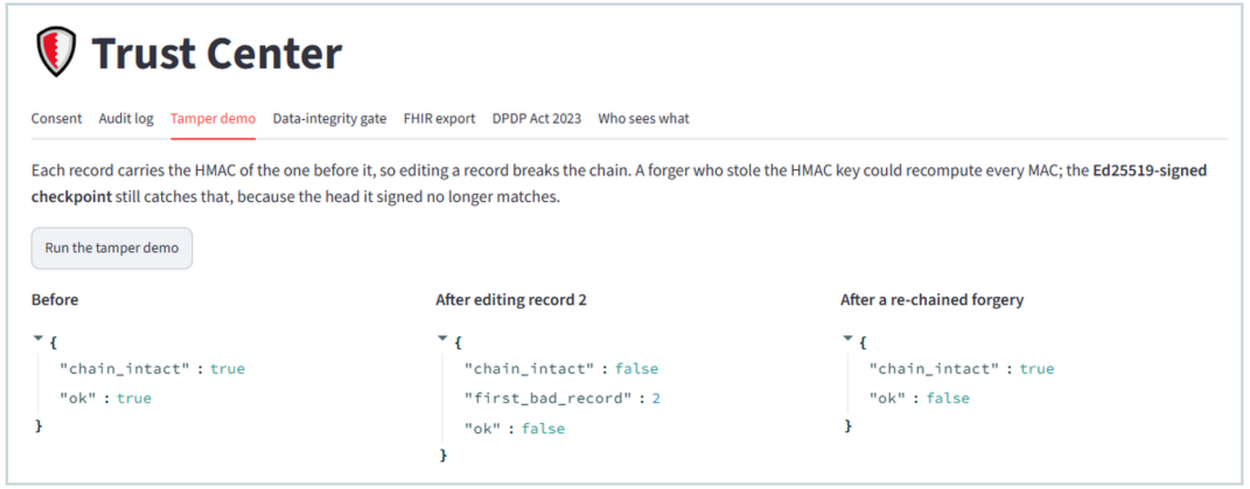
Possible heatstroke

Confusion, fainting, seizure, unresponsive or ≥40 °C → call 108 / 112, cool the person, nothing by mouth if not fully awake. No kidney score is shown.

Heatstroke definition and first aid follow India's National Action Plan on Heat Related Illnesses (NCDC/MoHFW). Two URGENT messages (no urine 12 h; cola-coloured urine with muscle pain) await clinician review of wording.

Trust Center: privacy you can demonstrate





ABDM-style consent

Ed25519-signed, checked on every read; revoke takes effect at once

6 roles, field-level

ASHA never sees genotype; lab sees only test orders

Tamper-evident audit

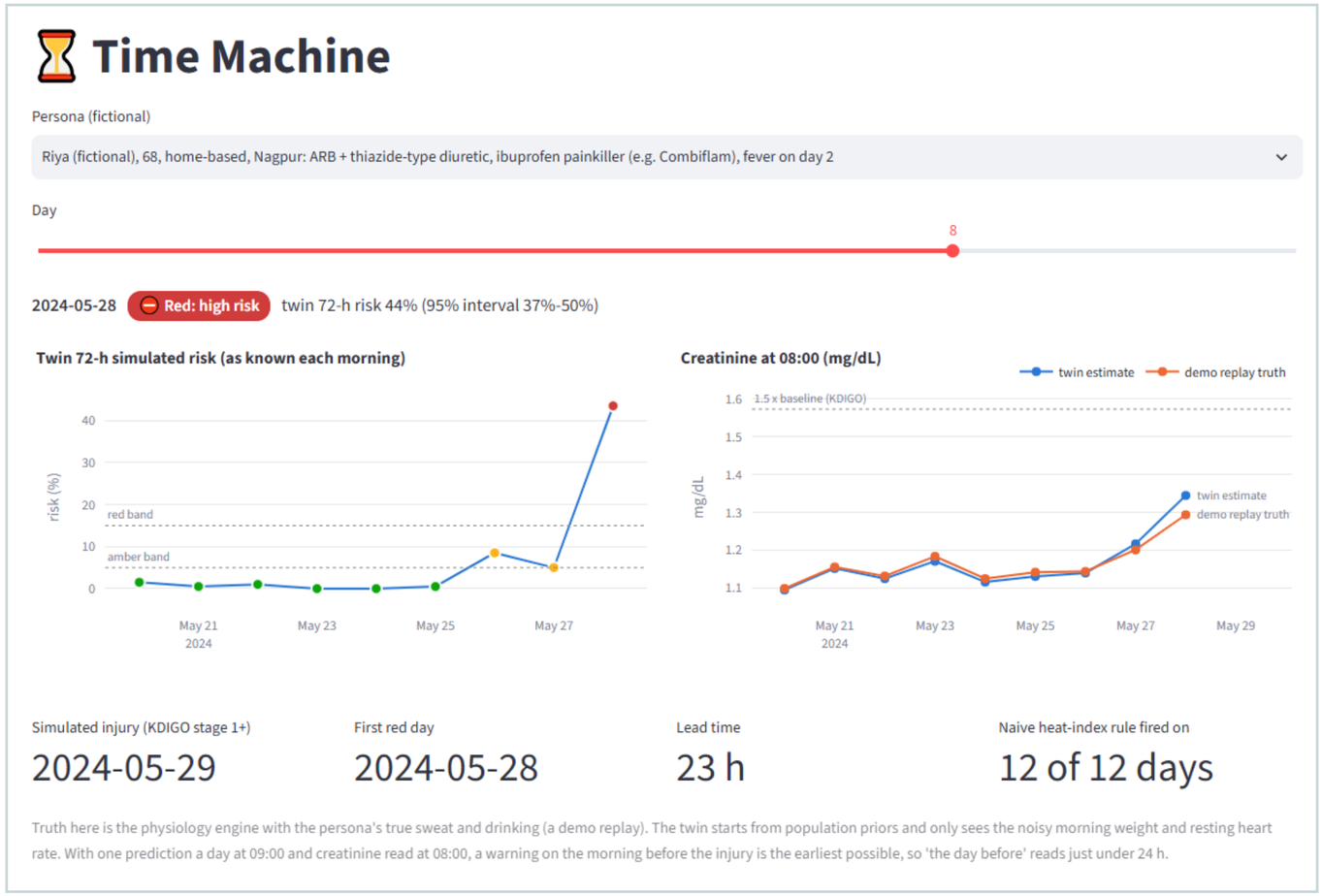
HMAC chain + signed checkpoints catch even a re-chained forgery

FHIR R4 + DPDP map

Standard export; 8 DPDP Act provisions mapped

Synthetic data only; no real patient data is used anywhere. Tested in tests/test_trust.py.

Time Machine: Riya's heatwave on real 2024 weather



23 h

warning before the simulated injury (red on 28 May, injury on 29 May)

12 of 12

days on which a simple heat-index rule fired for Riya, too often to act on

Green

all 12 days for her neighbour in the same heat, who drinks ORS and takes no painkiller

The twin starts from population priors and sees only noisy morning weight and heart rate; it never sees the simulated truth. Riya's fever began the day before the injury, so about a day is the earliest possible warning.

Evidence and results

How we built honest data, and what the numbers really say

Data: synthetic by necessity, checked against reality

Why synthetic

No open dataset links heat, wearables and medicines to daily creatinine for outpatients. The challenge rules allow synthetic data, so we built an enriched high-risk heat-season cohort.

10,000	training patients (2024 season)
2 × 5,000	held-out test patients (2025, 2026)
7	Indian cities on real hourly weather
164	parameters, each with a source (27 cited, 6 to verify, 131 assumptions)

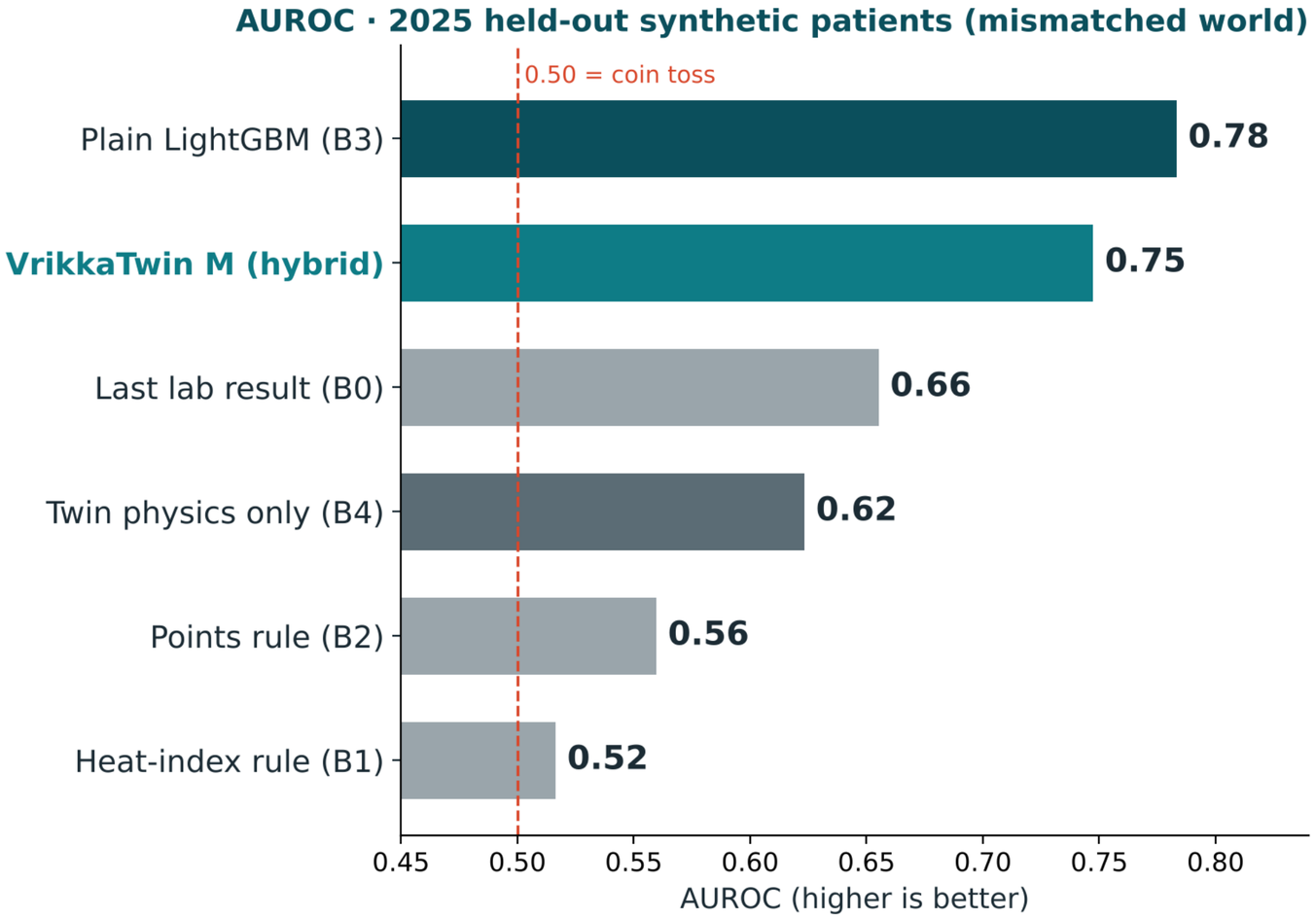
No circular grading

The 'truth' simulator is deliberately different from the twin: other kidney equations, hidden causes the twin cannot see (sepsis-like illness, muscle breakdown), wider patient variation. Headline results use this mismatched world.

Real open data used for checks

- MIMIC-IV & eICU demos: real creatinine, KDIGO labelling
- PMData: real Fitbit noise → heart-rate noise 4.04 bpm
- UCI CKD and Tawam (UAE, 491 patients): lab realism, slow clock
- Rejected: BD-KDD (implausible values), OSF data (no licence)

Results: the twin beats the rules clinics use today



54% of injuries got an amber or red warning (29% red)

18.7 alerts per injury caught, inside a budget of ≤ 0.5 red per 100 patient-days

0.90 AUROC when the world matches the twin (optimistic ceiling)

Synthetic (enriched high-risk heat-season cohort), mismatched simulator, 2025 held-out season, frozen model; prevalence 0.14% of days. Source: reports/metrics.json. 2026 season: M AUROC 0.73, plain LightGBM 0.77.

Honest scorecard: what passed, what did not

PASS · Labelling matches official MIMIC code

99.29% agreement (1,272 values)

PASS · No false red alerts on real healthy people

0 per 100 person-days (PMData, 1,845 days)

PASS · Triple-whammy effect in calibrated range

RR 2.3 vs Lapi 1.31-1.82 (calibration)

EQUAL · Slow-clock ranking (Tawam, real)

C-index 0.836 vs 0.834 baseline eGFR

MISSED · Lead-time target ≥ 24 h

23 h median: warnings come the day before

FAIL · Creatinine filter on ICU data

Carry-forward wins (0.53 vs 0.56 mg/dL)

FAIL · Cross-shift rises in cane cutters

Exertion & muscle effects under-modelled

LOSS · Hybrid vs plain LightGBM (AUPRC)

Plain wins on accuracy alone

Why we still lead with the twin: a plain classifier cannot simulate a decision, explain it physiologically, personalise from one person's scale, or say when to test.

What makes VrikkaTwin different

The empty 72-hour window

Kidney injury from heat + sickness + pills, between heat alarms (minutes) and kidney tests (years)

Decisions, not just scores

What-if levers with evidence badges and clinical guardrails

Genetics that changes the model

CYP2C9 changes ibuprofen exposure inside the physiology

Honest science

Mismatched simulator, pre-registered metrics, failures shown

Privacy that works

Consent on every read, signed audit, FHIR R4, DPDP map

Built for Bharat

4 languages, audio, brand-name pills, ASHA Tier 0, red flags first

Impact and next steps

How it could plug into Happiest Health, and what we would do next

How VrikkaTwin could plug into Happiest Health

Specialty clinics

An early-warning and triage list for endocrinology, personal-physician and future kidney patients

Diagnostics with home collection

The twin decides who needs a kidney test today and creates the home-collection task

Digital healthcare

Could serve as a 'kidney safety' module next to virtual consults: red alerts route to a doctor

Corporate wellness

A heat-season programme for field workforces: delivery, construction, facilities

"Proactive, not reactive": the twin turns a one-time test into a daily, personal safety net.

Limits we state openly, and the road ahead

Limitations

- Validated on synthetic patients; no real outpatient AKI labels exist openly
- Creatinine rises only after filtration falls; low muscle mass can hide injury in older people
- Dehydration alone cannot cause AKI in a healthy, drug-free twin (red flags cover severe cases)
- Heavy outdoor workers' short-term rises under-modelled
- Translations and two urgent messages await expert review · not a medical device

Roadmap

1. **Prospective pilot**
   A PHC or clinic with consented patients, weekly weights, routine creatinine, under ethics approval
2. **Real devices and ABDM**
   Band + scale pairing; ABDM sandbox consent; hosted deployment
3. **Clinical review**
   Nephrologist and pharmacist review of rules; native-speaker review
4. **New modules**
   Potassium, exertion physiology, transplant (CYP3A5)

Tech stack, licence and repository

Physiology & twin	pythermalcomfort 4.6.1 (ISO 7933) · filterpy 1.4.5 (UKF) · numpy · scipy · kfre 0.1.18
Machine learning	LightGBM 4.7.0 (monotone, isotonic calibration) · scikit-learn 1.9.1 · SHAP · lifelines
App & API	Streamlit 1.65.0 (8 role-based pages) · FastAPI · Plotly · gTTS audio · rapidfuzz brand matching
Security & interop	cryptography (Ed25519, Fernet) · HMAC audit chain · JWT · FHIR R4 · DuckDB (MIMIC checks)

Language

Python 3.12

Tests

218 passed (1 skipped)

Licence

MIT (code)

Folder

Monarch_Vellore-Institute-of-Technology

Repo

github.com/Eeshan842004/VrikkaTwin_Monarch

Data licences per source in data/README.md; restricted (ODbL) data is fetched by download scripts, never committed.

Thank you

VrikkaTwin protects the kidneys of people on BP pills, painkillers or with weak kidneys during Indian heatwaves and sick days, and tells the doctor which pill to review and when to test.



Research prototype · synthetic data and fictional personas · not a medical device

Harshita Kumar · Eeshan Gupta · Team Monarch, VIT Vellore

github.com/Eeshan842004/VrikkaTwin_Monarch